

HIGH AND RISING : A DECADE OF ANTIFUNGAL USE IN BELGIUM (2014-2023)

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Introduction

Emerging resistance in *Candida*, *Aspergillus*, and dermatophytes, monitoring antifungal use has become increasingly important. Belgium ranks 2nd highest consumer of antifungals and antimycotics for systemic use in Europe, nearly 3 times EU average (ECDC, 2023).

This study aims to assess national consumption trends between 2014 and 2023 across community and hospital settings.

Results

1. Community vs. Hospital use (2023)
 - Community: 96.2% oral use
 - Hospital: 3.8% of total use (60.2% oral, 39.8% parenteral)
2. Overall trend: Slight decline (-3.3%) between 2014 and 2023 but use remains high
3. Drug-level trends: Different ATC-5 drugs dominate in community vs. hospitals

Methods

- Settings Hospitals and community
- Drugs Reimbursed systemic antimycotics (ATC group J02) and antifungals (D01BA), based on ATC/DDD Index 2023
- Data sources
 - Consumption data, from National Institute for Health and Disability Insurance (reimbursement records)
 - Population data, from Eurostat
- Metric Defined daily doses (DDD)/1000 inhabitants/day (DID) Analysis Trends assessed using percentage relative change and Spearman's correlation

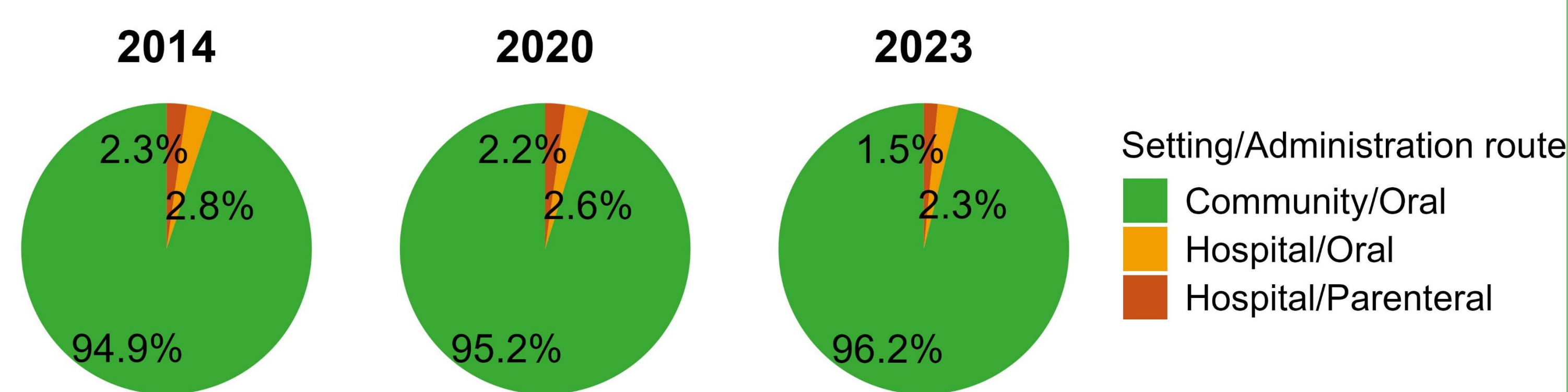


Fig. 1: Proportion of antifungal/antimycotic use in 2014, 2020 and 2023 by setting and administration route

Setting	Substance	2014 (in DID)	2023 (in DID)	% use by substance/setting (2023)	% relative change	Spearman correlation p-value
Total	J02 & D01BA	4.715	4.557		-3.3	0.04
Hospital	J02 & D01BA	0.240	0.175	100.0%	-27.3	< 0.01
	J02AC01 - fluconazole	0.153	0.096	55.2%	-37.1	< 0.01
	J02AC05 - isavuconazole	0.000	0.021	12.0%	13422.5	0.02
	J02AC04 - posaconazole	0.020	0.018	10.3%	-11.8	0.56
	J02AA01 - amphotericin B	0.026	0.015	8.8%	-41.0	0.03
	J02AC03 - voriconazole	0.018	0.010	5.7%	-46.0	0.71
	D01BA02 - terbinafine	0.008	0.006	3.7%	-15.2	0.05
	J02AC02 - itraconazole	0.009	0.004	2.0%	-61.3	< 0.01
	J02AX04 - caspofungin	0.004	0.003	1.5%	-28.9	0.61
	J02AX06 - anidulafungin	0.002	0.001	0.8%	-17.4	0.04
	J02AX01 - flucytosine	0.000	0.000	0.0%	-50.8	0.01
	J02AB02 - ketoconazole	0.000	0.000	0.0%	0.0	NA
Community	J02 & D01BA	4.474	4.382	100.0%	-2.1	0.09
	D01BA02 - terbinafine	1.823	2.149	49.0%	17.9	0.30
	J02AC01 - fluconazole	1.450	1.306	29.8%	-9.9	< 0.01
	J02AC02 - itraconazole	1.190	0.903	20.6%	-24.1	< 0.01
	J02AC03 - voriconazole	0.003	0.014	0.3%	414.5	0.66
	J02AC04 - posaconazole	0.009	0.009	0.2%	-6.8	0.12
	J02AB02 - ketoconazole	0.000	0.000	0.0%	0.0	NA

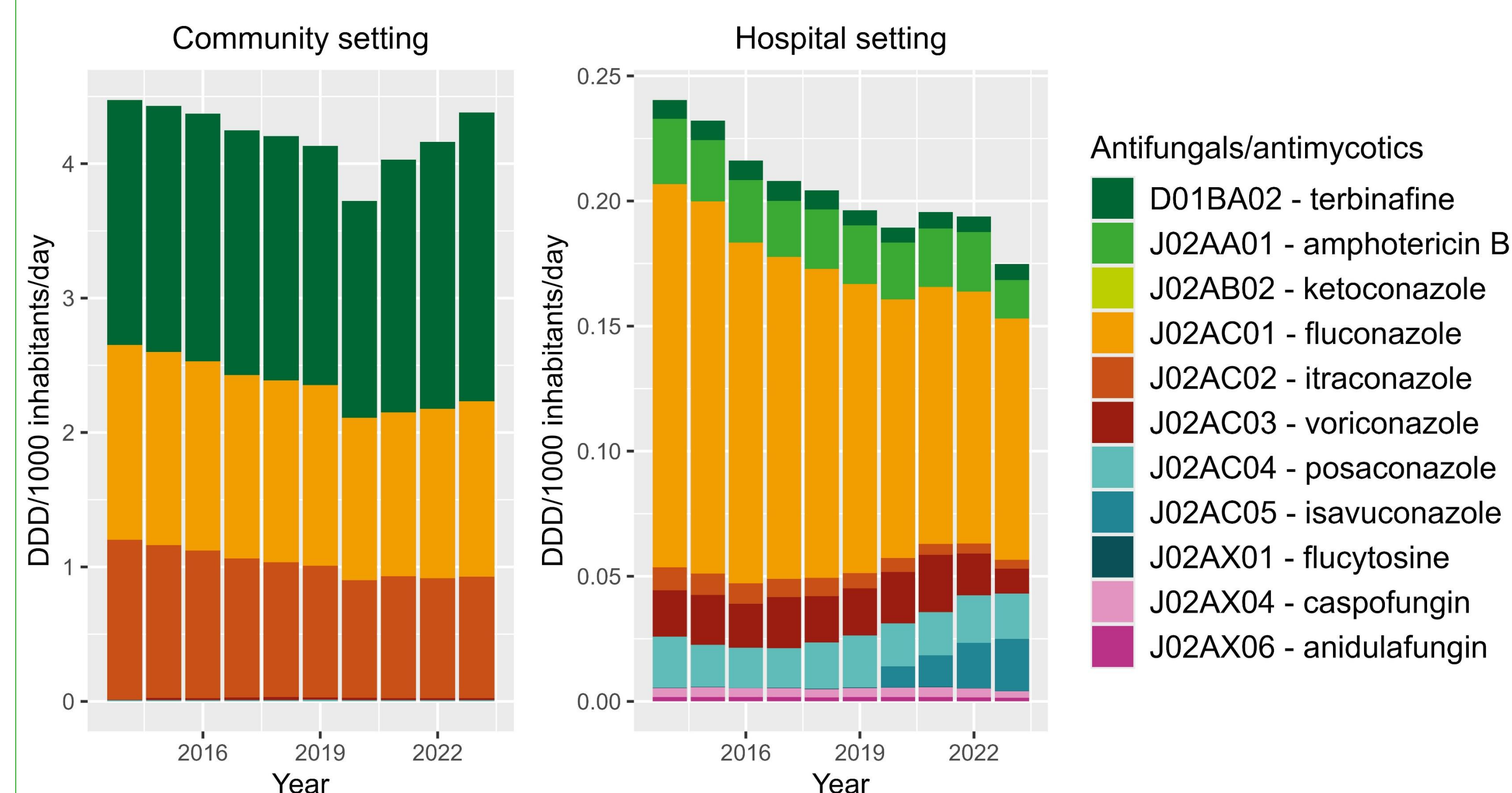


Fig. 2: Trends 2014-2023 in antifungal (D01BA) and antimycotic (J02) use at ATC-5 Level in community (left) and hospital (right) settings

Conclusion

- Antifungal consumption in Belgium remains high, mainly in the community
- Temporary drop during COVID-19, but post-2020 use is rising
- Continued reliance on azoles and terbinafine despite emerging resistance
- Implication: Need for reinforced antifungal stewardship

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